



Data Recorder (DR) Interface Control Document

Version 2.4.1

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Long Wavelength Array Project

Change Record

Version	Date	Affected Sections	Reason / Description
0.1	2009-09-12	All	Initial and second drafts; corrected typos, removed extraneous material.
0.2	2009-09-24	All	Third draft; restructured MIB, removed DFD command, added mechanical/electrical figures, use-case examples.
0.3	2009-10-04	All	Fourth draft; removed image matte, removed dev-stage diagnostics (except TST), revised electrical connections, updated formats.
0.4	2009-10-10	MIB, Commands	Final draft; removed EJT command and OP-ERROR MIB entry.
0.5	2010-01-28	–	Added BUF and SEL commands. Added DRSU-COUNT, DRSU-SELECTED, DRSU-INFO- $\{n\}$, BUFFER, BUFFER-RESTRICT MIB entries. Added NTP synchronization description. Fixed formatting.
1.0	2010-02-21	MIB, Commands	Fixed typos. UP command no longer returns TOTAL-STORAGE, no -F option. DWN faster. HDD-TEMP- $\{n\}$ updated for non-reporting hardware.
1.1	2010-07-02	§4, MIB, Commands	Partition/drive identification improved, added RAID volume support as external storage devices.
1.2	2010-08-25	MIB	Fixed errors in SCHEDULE-ENTRY- $\{n\}$ and DIRECTORY-ENTRY- $\{n\}$ MIB response formats.
1.3	2012-02-06	Commands	Added SPC command (spectrometer) and BUF update, removed language regarding deprecated data formats.
1.4	2012-03-02	Commands, §6	Changed spectrometer command to uniquely identify recordings. Fixed typos in spectrometer file format section.

Version	Date	Affected Sections	Reason / Description
2.4.1	2026-02-06	All	Major revision matching DROS version 2.4.1.0: • Reformatted to <code>lwa-icd.cls</code> template. • Replaced DP references with NDP throughout. • Removed TBN, TBW, and TBF data formats. • Added DRX8, TBT, TBS, and COR data format support. • Updated spectrometer documentation: multiple Stokes products (XXYY, CRCI, IQUV, IV, XXCRCIYY, I), flexible FFT length and integration count, Stokes metadata parameter. • Added multi-DR deployment documentation (<code>build_helper.py</code>, per-instance configuration). • Updated storage documentation: single drives preferred, RAID-0 arrays still supported for legacy DRSU compatibility. • Updated configuration parameters for <code>defaults_v2.cfg</code>. • Updated system requirements for modern Linux. • Added systemd service deployment documentation. • Added SHT, PNG, and RPT commands to command summary. • Added Appendix C (data format reference).

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Section 1

Description

1.1 Purpose

This document defines the interface between the Data Recorder (DR) subsystem and other Long Wavelength Array (LWA) station subsystems. The DR records the output of the Next-generation Digital Processor (NDP) subsystem and is controlled by the Monitor and Control System (MCS). Whereas station architecture and subsystem ICDs may refer to DR as a whole, this ICD applies to a single DR instance.

New in Version 2.4.1

In previous versions of this document, the upstream digital processor was referred to as DP (Digital Processing). All references have been updated to NDP (Next-generation Digital Processor) to reflect the current system architecture.

1.2 Scope

This ICD describes the DR's physical and electrical connections, software interfacing, and control methods. It covers:

- Physical and network connections between DR and other station subsystems.
- The MCS message-based command and monitoring interface.
- Configuration parameters and system requirements.
- Data format specifications for recorded and spectrometer output files.

1.3 System Overview

The DR runs the Data Recorder Operating Software version 2 (DROS2), which provides all ICD-specified functionality including MCS message processing, data recording, spectrometer computation, and storage management.

The build system compiles two DROS2 binaries from the same source code:

1. **Spectrometer mode** (primary): The DR records raw NDP data streams and can compute integrated power spectra of beam data in real time using the `SPC` command. This is the standard deployment mode.
2. **LiveBuffer mode** (secondary): The DR provides a live capture buffer for on-the-fly packet capture via the `BUF` command and `BUFFER` MIB entries. This mode is not the standard deployment.

The active mode is a runtime selection controlled by the contents of the `/LWA/config/binary-mode` file, which specifies either `Spectrometer` or `LiveBuffer`. The corresponding binary is launched when the DROS2 service starts; switching modes requires only updating this file and restarting the service. DROS2 is deployed as a `systemd` service (`dros.service`) that starts automatically on boot, restarts on failure, and logs to the system journal.

1.4 Multi-DR Deployment

New in Version 2.4.1

A station may deploy multiple DR instances on a single host, each recording from one or more NDP data streams. Multi-DR deployment is the preferred configuration and is managed through the `build_helper.py` utility, which creates per-instance build directories with separate:

- Installation paths (`/LWA/DRN/`)
- Storage mount points (`/LWA_STORAGE/DRN/`)
- Configuration files (`/LWA/DRN/config/`)
- Runtime directories (`/LWA/DRN/runtime/`)
- Service files (`dros-drN.service`)

Each DR instance has its own `defaults_v2.cfg` configuration file specifying a unique reference designator (e.g., `DR1`, `DR2`), IP address, and port numbers. When running in a single-DR configuration, the default installation paths (`/LWA/`, `/LWA_STORAGE/`) are used directly.

1.5 Document Organization

The remainder of this document is organized as follows:

Section 2 defines abbreviations, acronyms, and conventions used throughout this document.

Section 3 describes the physical and electrical interfaces of the DR.

Section 4 describes the monitor and control interface, including configuration, MIB structure, and control commands.

Section 5 describes the safety monitoring interface.

Section 6 lists references and related documents.

Appendix A defines the spectrometer data file format.

Appendix B lists error messages.

Appendix C provides a reference of all supported data formats.

Section 2

Document Conventions

2.1 Abbreviations and Acronyms

Abbreviation	Definition
AVX	Advanced Vector Extensions
COR	Correlator (data format)
DR	Data Recorder
DROS	Data Recorder Operating Software
DRX	Digital Receiver (4-bit data format)
DRX8	Digital Receiver 8-bit (data format)
DRSU	Data Recorder Storage Unit
EXT4	Fourth Extended File System
FFT	Fast Fourier Transform
FFTW	Fastest Fourier Transform in the West (FFT library)
GbE	Gigabit Ethernet
ICD	Interface Control Document
LWA	Long Wavelength Array
MCS	Monitor and Control System
MIB	Management Information Base
MJD	Modified Julian Day
MPM	Milliseconds Past Midnight
NDP	Next-generation Digital Processor
NTP	Network Time Protocol
RAID	Redundant Array of Independent Disks
SPC	Spectrometer Command
SSE	Streaming SIMD Extensions
TBS	Transient Buffer — Streaming (data format)
TBT	Transient Buffer — Triggered (data format)
UDP	User Datagram Protocol

2.2 Command Parameter Types

The following parameter types are used to describe command arguments and MIB response fields throughout this document:

uint8: Unsigned integer, 8 bits wide.

uint16: Unsigned integer, 16 bits wide.

uint32: Unsigned integer, 32 bits wide.

uint64: Unsigned integer, 64 bits wide.

ASCII-XXX-#: An ASCII string exactly *XXX* characters in length which is interpreted as a number. Valid characters are digits and right-padding spaces only.

ASCII-XXX-A: An ASCII string exactly *XXX* characters in length which is interpreted as a text string. Unless otherwise noted, valid characters are letters, numbers, the underscore character, and periods.

2.3 Mark-up Conventions

Symbol	Meaning	Example
<i>italics</i>	Variable or parameter name	<i>Start MPM</i>
fixed-width	Literal parameter or format value. A single-quote (') in a literal format denotes a space.	AB'_ = "A" + "B" + space + "_"
<...>	Variable substitution; brackets omitted from actual format	A_ = "A" + value of <i>B</i> + "_"
[...]	Optional parameter; brackets omitted from actual format	A[B]_ = "A" + optional <i>B</i> + "_"

2.4 Numeric Representation Convention

Numbers, units, and their associated prefixes and suffixes conform to the standard of IEC 60027-2. Specifically:

- Prefixes Ki, Mi, Gi, and Ti refer to 2^{10} , 2^{20} , 2^{30} , and 2^{40} , respectively.
- Prefixes K, M, G, and T refer to 10^3 , 10^6 , 10^9 , and 10^{12} , respectively.
- For binary size or rate: uppercase B represents a byte, lowercase b indicates a bit (e.g., MB = megabyte = 1,000,000 bytes; Kib = kibibit = 1,024 bits).

Section 3

Physical System Interfaces

3.1 Mechanical Interface

New in Version 2.4.1

The DR consists of a 1U rack-mount server (e.g., Gigabyte R152-Z30) with internal hot-swap drive bays for storage. No specific server model is required; any x86_64 system with sufficient CPU cores, RAM, and drive bays is suitable. The server is mounted in a standard EIA 19" rack.

3.2 Network Interfaces

The DR server connects to the station network with two Ethernet interfaces:

Interface	Speed	Purpose
Data interface	10 GbE (or faster)	Receives data frames from NDP. Connected to the NDP data network. The UDP port is configured by the <code>DataInPort</code> parameter in <code>defaults_v2.cfg</code> .
Control interface	1 GbE	Sends and receives MCS command/response messages. Connected to the station control network. The UDP ports are configured by the <code>MessageInPort</code> and <code>MessageOutPort</code> parameters in <code>defaults_v2.cfg</code> .

3.3 Storage

New in Version 2.4.1

The DR supports two storage configurations. Single-drive storage is preferred for new deployments; RAID-0 arrays remain supported for legacy DRSU compatibility.

3.3.1 Preferred: Single Drives

Individual hard drives are installed in internal hot-swap bays and formatted with the EXT4 file system. Each drive is mounted under the `/LWA_STORAGE/` hierarchy and managed independently by DROS2. This configuration simplifies drive replacement and eliminates the risk of array-wide data loss from a single drive failure.

3.3.2 Supported: RAID-0 Arrays (Legacy DRSU)

For compatibility with existing Data Recorder Storage Units (DRSUs), DROS2 also supports RAID-0 arrays constructed with `mdadm`. A DRSU typically consists of multiple drives combined into a single `/dev/md` device, formatted with EXT4, and mounted under `/LWA_STORAGE/`.

RAID-0 arrays provide higher sequential write bandwidth but offer no redundancy: failure of any constituent drive results in loss of all data on the array.

3.3.3 Storage Hierarchy

Regardless of configuration, all storage is mounted under `/LWA_STORAGE/` with the following structure:

```
/LWA_STORAGE/  
  Internal/  
    0/DROS/Rec/ # Raw recordings  
    0/DROS/Spec/ # Spectrometer output  
  External/  
    0/DROS/ # External storage (copy/dump targets)
```

In a multi-DR deployment (Section 1.4), each instance uses a separate storage root, e.g., `/LWA_STORAGE/DR1/`, `/LWA_STORAGE/DR2/`.

Storage setup (partitioning, formatting, and mounting) is managed by the `StorageControl.sh` script included with DROS2. DROS2 itself writes to mounted file systems and does not perform low-level storage management.

Section 4

Monitor and Control Interface

4.1 Overview

Control and monitoring of the DR is performed by two classes of messages:

1. **Monitoring messages** request system status information from the DR via the Management Information Base (MIB).
2. **Command messages** request that the DR execute some action.

All messages are exchanged over UDP using the format defined in the MCS Common ICD (LWA Engineering Memo MCS0005). If the DR cannot comply with a request, a rejection response containing an error message (see Appendix B) is returned.

4.2 System Configuration

DROS2 is configured through the `defaults_v2.cfg` file, read on startup or when manually reinitialized via the `INI` command (Section 4.8.1). The configuration parameters are described in Section 6.3.

New in Version 2.4.1

The configuration file has been renamed from `defaults.cfg` to `defaults_v2.cfg`. The `MessageOutURL` parameter has been removed; responses are sent to the originating IP address of each request.

In a multi-DR deployment (Section 1.4), each DR instance has its own configuration directory with a separate `defaults_v2.cfg` file.

4.3 Time Synchronization

The DR uses `ntpd` or `systemd-timesyncd` for time synchronization. The DR also synchronizes its internal clock to station time on each boot-up. The `SYN` command (Section 4.8.13) explicitly re-synchronizes time, as does the `INI` command (Section 4.8.1).

4.4 Timing Restrictions

- The DR supports up to 100 commands per second.
- Commands that schedule recording must allow at least 5 seconds between receipt and start of recording.
- Recordings may not be scheduled to begin within 5 seconds of the termination of a prior recording session.

4.5 MIB Structure

The Management Information Base is organized as shown in Table 4.1. Each entry is queried using the RPT command with the entry label as the data field.

Table 4.1: DR MIB Structure

Index	Label	Type	Description
1. MCS-Required Entries			
1.1	SUMMARY	Status	Current system status
1.2	INFO	Status	Detailed status information
1.3	LASTLOG	Status	Last log entry
1.4	SUBSYSTEM	Status	Subsystem identifier
1.5	SERIALNO	Status	Serial number
1.6	VERSION	Status	Software version
2. Current Operation			
2.1	OP-TYPE	Status	Current operation type
2.2	OP-SCHEDULE	Status	Operation start and stop times
2.2.1	OP-START	Status	Operation start time
2.2.2	OP-STOP	Status	Operation expected stop time
2.3	OP-REFERENCE	Status	Command reference number
2.4.1	OP-TAG	Status	Internal file tag
2.4.2	OP-FORMAT	Status	Data format in use
2.4.3	OP-POSITION	Status	File position info
2.5.1	OP-FILENAME	Status	External file name and device
2.5.2	OP-FILEINDEX	Status	File series index
3. Schedule			
3.1	SCHEDULE-COUNT	Status	Count of scheduled operations
3.2.{n}	SCHEDULE-ENTRY-{n}	Status	{n}th scheduled operation
4. Directory			
4.1	DIRECTORY-COUNT	Status	Count of stored recordings
4.2.{n}	DIRECTORY-ENTRY-{n}	Status	{n}th recording entry
5. Storage			
5.1	TOTAL-STORAGE	Status	Total internal storage (bytes)
5.2	REMAINING-STORAGE	Status	Available internal storage (bytes)
5.3	CONTIGUOUS-STORAGE	Status	Deprecated; always returns 0

Index	Label	Type	Description
6. Removable Devices			
6.1	DEVICE-COUNT	Status	Number of external devices
6.2.{n}	DEVICE-ID-{n}	Status	Device ID of {n}th device
6.3.{n}	DEVICE-STORAGE-{n}	Status	Free space on {n}th device
6.4.{n}	DEVICE-BARCODE-{n}	Status	Volume name of {n}th device
7. CPU Info			
7.1	CPU-COUNT	Status	Number of CPU cores
7.2.{n}	CPU-TEMP-{n}	Status	Temperature of core {n}
8. HDD Info			
8.1	HDD-COUNT	Status	Number of hard drives
8.2.{n}	HDD-TEMP-{n}	Status	Temperature of HDD {n}
9. Data Formats			
9.1	FORMAT-COUNT	Status	Number of supported formats
9.2.{n}	FORMAT-NAME-{n}	Status	Name of {n}th format
9.3.{n}	FORMAT-PAYLOAD-{n}	Status	UDP payload size of {n}th format
9.4.{n}	FORMAT-RATE-{n}	Status	Data rate of {n}th format
10. System Log			
10.1	LOG-COUNT	Status	Number of log entries
10.2.{n}	LOG-ENTRY-{n}	Status	{n}th log entry
11. Live Buffer			
11.1	BUFFER	Data	Deprecated; see Section 4.6.38
11.2	BUFFER-RESTRICT	Data	Deprecated; see Section 4.6.39
12. DRSU Status			
12.1	DRSU-COUNT	Status	Number of detected DRSUs
12.2	DRSU-SELECTED	Status	Currently selected DRSU
12.3	DRSU-BARCODE	Status	Barcode of selected DRSU
12.4.{n}	DRSU-INFO-{n}	Status	Info on {n}th detected DRSU
13. Timing			
13.1	TT-LAG	Status	Current time-tag lag
13.2	TT-LAG-INITIAL	Status	Initial time-tag lag

4.6 MIB Entries in Detail

4.6.1 SUMMARY

Index:

1.1

Description:

Reports the current system status.

Response Format:

<Status>

Response Element:

Status (ASCII-7-A): One of the values in Section 4.9.

4.6.2 INFO**Index:**

1.2

Description:

Reports detailed status information. The content varies depending on the current system state.

Response Format:

<Info>

Response Element:

Info (ASCII-256-A): Human-readable status detail.

4.6.3 LASTLOG**Index:**

1.3

Description:

Reports the most recent system log entry.

Response Format:

<Log Entry>

Response Element:

Log Entry (ASCII-256-A): Last log message.

4.6.4 SUBSYSTEM**Index:**

1.4

Description:

Reports the subsystem reference designator.

Response Format:

<Designator>

Response Element:

Designator (ASCII-3-A): Subsystem identifier (e.g., DR1).

4.6.5 SERIALNO**Index:**

1.5

Description:

Reports the DR serial number as configured in `defaults_v2.cfg`.

Response Format:

<Serial Number>

Response Element:

Serial Number (ASCII-20-A).

4.6.6 VERSION**Index:**

1.6

Description:

Reports the DROS2 software version.

Response Format:

<Version>

Response Element:

Version (ASCII-256-A): Software version string.

4.6.7 OP-TYPE**Index:**

2.1

Description:

Reports the current operation type. If no operation is in progress, reports idle state.

Response Format:

<Operation Type>

Response Element:

Operation Type (ASCII-11-A) — one of the values in Table 4.2.

Table 4.2: Operation Type Values

Value	Meaning
Idle	No operation in progress
Record	Recording NDP data
Spectrometr	Computing spectrometer output
Copy	Copying recording to external storage
Dump	Dumping blocks to external storage
Format	Formatting storage device

4.6.8 OP-SCHEDULE**Index:**

2.2

Description:

Reports the start and stop times of the current operation as a single combined response. Not valid if operation is Idle.

Response Format:

<Start MJD>'<Start MPM>'<Stop MJD>'<Stop MPM>

Response Elements:

Start MJD (ASCII-6-#): MJD when operation began.

Start MPM (ASCII-9-#): MPM when operation began.

Stop MJD (ASCII-6-#): MJD when operation will end.

Stop MPM (ASCII-9-#): MPM when operation will end.

4.6.9 OP-START**Index:**

2.2.1

Description:

Reports the time at which the current operation began. Not valid if operation is **Idle**.

Response Format:

<Start MJD>'<Start MPM>

Response Elements:

Start MJD (ASCII-6-#): MJD when operation began.

Start MPM (ASCII-9-#): MPM when operation began.

4.6.10 OP-STOP**Index:**

2.2.2

Description:

Reports the scheduled/expected end time of the current operation. Only valid for **Record**, **Spectrometr**, **Copy**, or **Dump** operations.

Response Format:

<Stop MJD>'<Stop MPM>

Response Elements:

Stop MJD (ASCII-6-#): MJD when operation will end.

Stop MPM (ASCII-9-#): MPM when operation will end.

4.6.11 OP-REFERENCE**Index:**

2.3

Description:

Reports the reference number of the command message that initiated/scheduled the current operation. Not valid if operation is **Idle**.

Response Format:

<Reference Number>

Response Element:

Reference Number (ASCII-9-#).

4.6.12 OP-TAG**Index:**

2.4.1

Description:

Reports the tag value identifying the file in use. Only valid for **Record**, **Spectrometr**, **Copy**, or **Dump** operations.

Response Format:

<Tag>

Response Element:

Tag (ASCII-16-A): filename tag in the form <MJD>_<Reference Number> including the literal underscore.

4.6.13 OP-FORMAT**Index:**

2.4.2

Description:

Reports the data format in use. For **Record** operations, this is the format being recorded. For **Copy/Dump**, it is the format specified by the originating **REC** command.

Response Format:

<Data Format>

Response Element:

Data Format (ASCII-32-A).

4.6.14 OP-POSITION**Index:**

2.4.3

Description:

Reports start position, length, and current position of the file in use. Current Position is the offset relative to Start Position. Valid for **Record**, **Copy**, **Dump** operations. In **Spectrometr** mode, Start Position and Length are not valid, but Current Position reflects the current output file size.

Response Format:

<Start Position>'<Length>'<Current Position>

Response Elements:

Start Position (ASCII-15-#): Position of first byte (always 0 for Recording).

Length (ASCII-15-#): Copy: bytes to copy; Dump: chunk size; Record: expected file size.
Current Position (ASCII-15-#): Position of most recent byte written/read.

4.6.15 OP-FILENAME

Index:

2.5.1

Description:

Reports the file name and external storage device ID. Only valid for Copy or Dump operations. For Dump, the returned file name is the series name; individual files are named <Filename>.X with a zero-padded serial identifier.

Response Format:

<Storage ID>'<Filename>

Response Elements:

Storage ID (ASCII-64-A): Linux partition or mount point.

Filename (ASCII-128-A): File or series name.

4.6.16 OP-FILEINDEX

Index:

2.5.2

Description:

Reports which file of a file series is being written to. Only valid for Dump operations.

Response Format:

<File Index>

Response Element:

File Index (ASCII-9-#).

4.6.17 SCHEDULE-COUNT

Index:

3.1

Description:

Reports the count of all scheduled operations, including the current operation if one is in progress.

Response Format:

<Count>

Response Element:

Count (ASCII-6-#).

4.6.18 SCHEDULE-ENTRY-{n}

Index:

3.2.{n}

Description:

Reports relevant information for the {n}th scheduled operation.

Response Format:

<Operation Type>'<Reference Number>'<Start MJD>'<Start MPM>'<Stop MJD>'<Stop MPM>'<Data Format>

Response Elements:

Operation Type (ASCII-11-A).
 Reference Number (ASCII-9-#).
 Start MJD (ASCII-6-#).
 Start MPM (ASCII-9-#).
 Stop MJD (ASCII-6-#).
 Stop MPM (ASCII-9-#).
 Data Format (ASCII-32-A).

4.6.19 DIRECTORY-COUNT**Index:**

4.1

Description:

Reports the number of recordings contained on internal storage.

Response Format:

<Count>

Response Element:

Count (ASCII-6-#).

4.6.20 DIRECTORY-ENTRY-{n}**Index:**

4.2.{n}

Description:

Reports pertinent information for the {n}th recording on internal storage.

Response Format:

<Tag>'<Start MJD>'<Start MPM>'<Stop MJD>'<Stop MPM>'<Data Format>'<Size>'<Disk Usage>'<Complete>

Response Elements:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.
 Start MJD (ASCII-6-#).
 Start MPM (ASCII-9-#).
 Stop MJD (ASCII-6-#).
 Stop MPM (ASCII-9-#).
 Data Format (ASCII-32-A): Format used when recorded.
 Size (ASCII-15-#): Bytes written to disk.

Disk Usage (ASCII-15-#): Total bytes occupied on disk.
Complete (ASCII-3-A): YES or NO .

4.6.21 TOTAL-STORAGE

Index:

5.1

Description:

Reports total storage capacity of internal storage in bytes. Returns 0 when storage is offline or unavailable.

Response Format:

<Size>

Response Element:

Size (ASCII-15-#): Total internal storage in bytes.

4.6.22 REMAINING-STORAGE

Index:

5.2

Description:

Reports available bytes on internal storage.

Response Format:

<Available>

Response Element:

Available (ASCII-15-#): Unused internal storage in bytes.

4.6.23 CONTIGUOUS-STORAGE

Index:

5.3

Description:

Deprecated. Always returns 0. Retained for backward compatibility with older MCS implementations.

Response Format:

<Size>

Response Element:

Size (ASCII-15-#): Always 0.

4.6.24 DEVICE-COUNT

Index:

6.1

Description:

Reports the number of available external storage devices.

Response Format:

<Count>

Response Element:

Count (ASCII-6-#).

4.6.25 DEVICE-ID- $\{n\}$ **Index:**

6.2. $\{n\}$

Description:

Reports the device ID of the $\{n\}$ th external storage device.

Response Format:

<Storage ID>

Response Element:

Storage ID (ASCII-64-A): Linux partition (e.g., `/dev/sdf1`).

4.6.26 DEVICE-STORAGE- $\{n\}$ **Index:**

6.3. $\{n\}$

Description:

Reports free storage space on the $\{n\}$ th external storage device.

Response Format:

<Available>

Response Element:

Available (ASCII-15-#): Unused bytes. Returns 0 if the device is not properly formatted or uses an unsupported file system.

4.6.27 DEVICE-BARCODE- $\{n\}$ **Index:**

6.4. $\{n\}$

Description:

Reports the volume name (barcode) of the $\{n\}$ th external storage device.

Response Format:

<Volume Name>

Response Element:

Volume Name (ASCII-64-A): Volume label of the device.

4.6.28 CPU-COUNT**Index:**

7.1

Description:

Reports the number of CPU cores.

Response Format:

<Count>

Response Element:

Count (ASCII-3-#).

4.6.29 CPU-TEMP-{n}**Index:**

7.2.{n}

Description:

Reports the temperature of CPU core {n}.

Response Format:

<Core {n} Temp>

Response Element:

Core {n} Temp (ASCII-3-#): Temperature in degrees Celsius.

4.6.30 HDD-COUNT**Index:**

8.1

Description:

Reports the number of hard drives comprising internal storage.

Response Format:

<Count>

Response Element:

Count (ASCII-3-#).

4.6.31 HDD-TEMP-{n}**Index:**

8.2.{n}

Description:

Reports the temperature of the {n}th hard drive. Some hardware may not support temperature reporting; in such cases the response may be empty.

Response Format:

<HDD {n} Temp>

Response Element:

HDD {n} Temp (ASCII-3-#): Temperature in degrees Celsius.

4.6.32 FORMAT-COUNT

Index:

9.1

Description:

Returns the number of recording formats supported.

Response Format:

<Count>

Response Element:

Count (ASCII-6-#).

4.6.33 FORMAT-NAME-**{n}**

Index:9.2.**{n}****Description:**Returns the name of the **{n}**th recording format.**Response Format:**

<Format Name>

Response Element:

Format Name (ASCII-32-A).

4.6.34 FORMAT-PAYLOAD-**{n}**

Index:9.3.**{n}****Description:**Returns the UDP packet payload size of the **{n}**th recording format.**Response Format:**

<UDP Payload Size>

Response Element:

UDP Payload Size (ASCII-4-#): Bytes of payload.

4.6.35 FORMAT-RATE-**{n}**

Index:9.4.**{n}****Description:**Returns the data rate of the **{n}**th recording format in bytes per second.**Response Format:**

<Rate>

Response Element:

Rate (ASCII-9-#): Overall data rate (MAC/UDP headers excluded).

4.6.36 LOG-COUNT

Index:

10.1

Description:

Reports the number of system log entries.

Response Format:

<Count>

Response Element:

Count (ASCII-6-#).

4.6.37 LOG-ENTRY-{n}

Index:

10.2.{n}

Description:

Reports the {n}th entry in the system log.

Response Format:

<MJD>'<MPM>'<Message Class>'<Message>

Response Elements:

MJD (ASCII-6-#): MJD when entry was logged.

MPM (ASCII-9-#): MPM when entry was logged.

Message Class (ASCII-7-A): **info**, **warning**, or **error** (trailing spaces pad to 7 characters).

Message (ASCII-234-A): Human-readable description (≤ 234 characters).

4.6.38 BUFFER

Index:

11.1

Description:

Deprecated. Returns a deprecation notice. This entry was previously used to return the contents of the live capture buffer.

Deprecation Notice

The BUFFER MIB entry is deprecated in the current DROS2 implementation. Queries will receive a rejection response.

4.6.39 BUFFER-RESTRICT

Index:

11.2

Description:

Deprecated. Returns a deprecation notice. This entry was previously used to return live capture buffer contents limited to one data packet.

Deprecation Notice

The BUFFER-RESTRICT MIB entry is deprecated in the current DROS2 implementation. Queries will receive a rejection response.

4.6.40 DRSU-COUNT**Index:**

12.1

Description:

Reports the number of detected DRSUs.

Response Format:

<Count>

Response Element:

Count (ASCII-2-#).

4.6.41 DRSU-SELECTED**Index:**

12.2

Description:

Reports which DRSU is currently active. If internal storage is down, reflects the last valid selection.

Response Format:

<DRSU Number>

Response Element:

DRSU Number (ASCII-2-#).

4.6.42 DRSU-BARCODE**Index:**

12.3

Description:

Reports the barcode (volume name) of the currently selected DRSU.

Response Format:

<Barcode>

Response Element:

Barcode (ASCII-64-A): Volume label of the selected DRSU.

4.6.43 DRSU-INFO- $\{n\}$

Index:

12.4. $\{n\}$

Description:

Reports information on the $\{n\}$ th detected DRSU.

Response Format:

<Name>'<Partition>'<Unformatted Size>

Response Elements:

Name (ASCII-6-A): DRSU name (e.g., DRSU00).

Partition (ASCII-64-A): Partition on which the DRSU resides.

Unformatted Size (ASCII-16-#): Size in bytes before formatting.

4.6.44 TT-LAG

Index:

13.1

Description:

Reports the current time-tag lag of the data receiver. The lag is the difference between the expected and observed time tags in the incoming NDP data stream, expressed in NDP clock ticks.

Response Format:

<Lag>

Response Element:

Lag (ASCII-15-#): Current time-tag lag in clock ticks.

4.6.45 TT-LAG-INITIAL

Index:

13.2

Description:

Reports the time-tag lag measured at the start of the most recent recording session.

Response Format:

<Lag>

Response Element:

Lag (ASCII-15-#): Initial time-tag lag in clock ticks.

4.7 Control Commands Summary

Table 4.3 lists the control commands supported by the DR.

Table 4.3: DR Control Commands

Command	Description	Section
INI	Initialize / restore to boot-up state	4.8.1
REC	Schedule a recording	4.8.2
DEL	Delete a recording	4.8.3
STP	Stop a scheduled or in-progress operation	4.8.4
GET	Retrieve portion of a recording	4.8.5
CPY	Copy recording to external storage	4.8.6
DMP	Dump recording to file series	4.8.7
FMT	Format storage device	4.8.8
DWN	Bring internal storage offline	4.8.9
UP	Bring internal storage online	4.8.10
SEL	Select DRSU	4.8.11
BUF	Set up live capture buffer	4.8.12
SYN	Synchronize with NTP server	4.8.13
SPC	Schedule spectrometer recording	4.8.14
SHT	Shut down DROS2	4.8.15
PNG	Ping (status query)	4.8.16
RPT	Report (MIB query)	4.8.17
TST	Self-test (development only)	4.8.18

4.8 Control Commands in Detail

For each command below, the argument format and response format are described. If no arguments are specified, none are required. If no response format is described, the R-COMMENT field is empty on success. Failed commands return R in R-RESPONSE with an error message in R-COMMENT (see Appendix B).

4.8.1 INI — Initialize

Description:

Restores the DR to its initial boot-up state except for the system log and internal storage contents.

Argument Format:

<Flags>

Argument:

Flags (ASCII-256-A): Specify `--flush-log` or `-L` to force system log re-initialization; `--flush-data` or `-D` to force internal storage re-initialization. The field need not be padded; flag order does not matter.

4.8.2 REC — Record

Description:

Schedules or initiates recording of NDP data. Upon success, returns a tag value uniquely identifying the recorded file.

Argument Format:

<Start MJD>'<Start MPM>'<Length>'<Data Format>

Arguments:

Start MJD (ASCII-6-#): Modified Julian Day; not > 24 hours in the future.

Start MPM (ASCII-9-#): Milliseconds past midnight; not within 5 s of a prior operation's termination.

Length (ASCII-9-#): Duration in milliseconds.

Data Format (ASCII-32-A): Pre-configured format name (see Section 4.6.33 and Appendix C).

Response Format:

<Tag>

Response Element:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

New in Version 2.4.1

The supported data format names have been updated. The following formats are available: DEFAULT_DRX, DRX_FILT_1 through DRX_FILT_7, DEFAULT_DRX8, DRX8_FILT_1 through DRX8_FILT_7, DEFAULT_TBT, DEFAULT_TBS, TBS_FILT_7 through TBS_FILT_9, DEFAULT_COR. The previously supported DEFAULT_TBN and DEFAULT_TBW formats have been removed. See Appendix C for details.

Note

Example: To schedule a 30-second DRX recording starting at MJD 60700, MPM 43200000 with reference number 42:

REC 060700 043200000 000030000 DEFAULT_DRX

Response: 060700_000000042

4.8.3 DEL — Delete**Description:**

Deletes a recording from internal storage.

Argument Format:

<Tag>

Argument:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

4.8.4 STP — Stop**Description:**

Halts or prevents a specified operation. If the operation is scheduled but not yet in progress, it is deleted from the schedule and disk space is freed. If it is in progress, it is halted and the file is closed but not deleted. Also stops scheduled or in-progress spectrometer operations.

Argument Format:

<Tag>

Argument:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

4.8.5 GET — Get**Description:**

Retrieves a portion of a specified recording.

Argument Format:

<Tag>'<Start Byte>'<Length>

Arguments:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

Start Byte (ASCII-15-#): Byte offset within the file.

Length (ASCII-15-#): Bytes to return; limited to 8146 bytes.

Response Format:

<Data>

Response Element:

Data (uint8)×Length: Requested bytes from the file.

4.8.6 CPY — Copy**Description:**

Copies a portion of a recording to a file on external storage. Existing files are overwritten without warning. Not available if recordings are scheduled.

Argument Format:

<Tag>'<Start Byte>'<Length>'<Device ID>'<Filename>

Arguments:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

Start Byte (ASCII-15-#): Byte offset within the file.

Length (ASCII-15-#): Bytes to copy.

Storage ID (ASCII-64-A): Target device partition.

Filename (ASCII-128-A): Destination file name.

4.8.7 DMP — Dump**Description:**

Copies data blocks from a recording to a series of files on external storage. Files are named <Filename>.X with a zero-padded serial identifier. Not available if recordings are scheduled.

Argument Format:

<Tag>'<Start Byte>'<Length>'<Block Size>'<Device ID>'<Filename>

Arguments:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.
Start Byte (ASCII-15-#): Byte offset within the file.
Length (ASCII-15-#): Bytes to copy.
Block Size (ASCII-15-#): Bytes per output file.
Storage ID (ASCII-64-A): Target device partition.
Filename (ASCII-128-A): Series name.

4.8.8 FMT — Format**Description:**

Formats an internal or external storage device. This is a destructive operation that erases all data on the target device. External formatting may require substantial time proportional to device size.

Argument Format:

[Storage ID]

Argument:

Storage ID (ASCII-64-A): Optional external device partition. If omitted, internal storage is formatted.

4.8.9 DWN — Down**Description:**

Prepares internal storage for removal or replacement. Executes immediately; requires a few seconds for completion. Wait at least 1 minute after DWN before physically disconnecting storage.

4.8.10 UP — Up**Description:**

Scans for internal storage and brings it online. If file system information cannot be determined, the command is rejected.

4.8.11 SEL — Select DRSU**Description:**

Selects an alternate DRSU for internal storage. The DR must be idle with no scheduled operations. DRSUs must have been previously detected by DROS2 (see `DRSU-INFO-{n}`).

Argument Format:

<DRSU Identifier>

Argument:

DRSU Identifier: Either a 0-based numeric index (DRSU to activate; 0 = first, 1 = second, etc.) or a DRSU barcode string matching the volume name reported by `DRSU-BARCODE`.

4.8.12 BUF — Live Capture Buffer Setup

Description:

Initializes the live capture buffer parameters for the next recording start. The buffer stores up to 100 ms of data and can trigger once or periodically. Parameters are reused with subsequent recordings. Buffer contents are overwritten unless retrieved via the BUFFER or BUFFER-RESTRICT MIB entries; if retrieved, contents are held and subsequent captures go to a shadow buffer.

Argument Format:

<OffsetType><WidthType><Retrigger><Order><Behaviour>'<Offset>'<Width>'<Interval>

Arguments:

OffsetType (ASCII-1-A): T = milliseconds, I = packet counts (applies to Offset and Interval).

WidthType (ASCII-1-A): T = milliseconds, I = packet counts (applies to Width).

Retrigger (ASCII-1-A): A = periodic, N = one-shot.

Order (ASCII-1-A): Deprecated; supply A or B (ignored).

Behaviour (ASCII-1-A): D = delete/reuse, H = hold/reuse, R = hold/no-reuse.

Offset (ASCII-16-#): Milliseconds or packets from recording start.

Width (ASCII-16-#): Capture length in milliseconds or packets.

Interval (ASCII-16-#): Milliseconds or packets between captures (when Retrigger = A); should be much greater than Width.

Implementation Note

The BUF command is only functional when DROS2 is built in LiveBuffer mode.

4.8.13 SYN — Synchronize

Description:

Explicitly synchronizes the DR clock with the station NTP time server. Executing while operations are scheduled or in progress may result in recording more or less data than desired.

4.8.14 SPC — Spectrometer

Description:

Schedules an integrated spectra recording session. The spectrometer computes power spectra for each tuning and polarization of beam data (NDP DRX or DRX8 output) without windowing. Non-beam data formats (TBT, TBS, COR) are not supported by the spectrometer. Output is stored on the internal storage under the spectrometer directory. The tag value uniquely identifies the output file. The file format is defined in Appendix A.

Argument Format:

<Start MJD>'<Start MPM>'<Length>'<Transform Length>'<Integration Count>' [Minimum Fill] [Highwater Threshold]

Arguments:

Start MJD (ASCII-6-#): Modified Julian Day; not > 24 hours in the future.

Start MPM (ASCII-9-#): Milliseconds past midnight.

Length (ASCII-9-#): Duration in milliseconds. Same restrictions as REC.

Transform Length (ASCII-6-#): FFT length in channels. Valid range: 32–16384.

Integration Count (ASCII-8-#): Number of consecutive spectra to integrate. Valid range: 384–24576.

Minimum Fill (ASCII-8-#): Optional (default = Integration Count / 2). Minimum number of samples in a “cell” for valid output. Range: 2 to Integration Count.

Highwater Threshold (ASCII-4-#): Optional (default = 128). Minimum buffered “blocks” before partial processing.

Validation:

The product (Transform Length × Integration Count) must be an integral multiple of 4096 (the DRX samples-per-frame count).

Response Format:

<Tag>

Response Element:

Tag (ASCII-16-A): Form <MJD>_<Reference Number>.

Implementation Note

The SPC command is only available when DROS2 is built in Spectrometer mode.

New in Version 2.4.1

The spectrometer has been significantly enhanced since version 1.4:

- **Stokes products:** The output Stokes product is selected via the **Stokes** metadata field in the MCS message. Supported values: **XXYY** (default), **CRCI**, **IQUV**, **IV**, **XXCRCIYY**, **I**.
- **Flexible FFT parameters:** The Transform Length and Integration Count are now fully configurable within the supported ranges (32–16384 channels, 384–24576 integrations). The previous restriction to 32 channels and 6144 integrations has been removed.
- **DRX and DRX8 input:** The spectrometer accepts both DRX (4-bit) and DRX8 (8-bit) input data.

Note

Example: To schedule a 60-second spectrometer session at MJD 60700, MPM 43200000 with 256 channels, 4096 integrations, and XXYY Stokes output:

SPC 060700 043200000 000060000 000256 00004096

Response: 060700_000000042

4.8.15 SHT — Shutdown

Description:

Initiates a shutdown of DROS2. The system state transitions to **SHUTDWN** and all resources are released. When deployed as a **systemd** service, the process will be restarted automatically after the configured delay.

Argument Format:

[Mode]

Argument:

Mode (ASCII-7-A): Optional. **SCRAM** for an immediate shutdown that bypasses normal cleanup. **RESTART** to request a clean restart. If omitted, an orderly shutdown is performed.

4.8.16 PNG — Ping**Description:**

Returns the current system state in the R-COMMENT field. Used as a simple connectivity and status check.

Response Format:

<System State>

Response Element:

System State (ASCII-7-A): One of the values in Section 4.9.

4.8.17 RPT — Report**Description:**

Queries a MIB entry and returns its value. The MIB entry label is supplied as the data field of the command message.

Argument Format:

<MIB Label>

Argument:

MIB Label: The label of the MIB entry to query (see Section 4.5).

Response:

The response data field contains the MIB entry value as described in Section 4.6.

4.8.18 TST — Self Test**Description:**

Performs a system self-test. This command is for development purposes only and is not supported by this ICD.

4.9 System States

The DR reports one of the following system states in response to PNG commands and in the R-SUMMARY field of all responses:

State	Description
SHUTDWN	The DR is in the process of shutting down.
BOOTING	The DR is initializing (scanning storage, starting services).
NORMAL	The DR is operating normally and ready to accept commands.

WARNING The DR has detected a non-critical condition (e.g., elevated temperature, low storage space).

ERROR The DR has detected a critical error preventing normal operation.

Section 5

Safety Interface

The DR is primarily a software system running on standard PC hardware and does not present significant physical safety hazards beyond those associated with normal computing equipment. The following monitoring and protection mechanisms are provided.

5.1 Temperature Monitoring

CPU core temperatures are monitored via the `CPU-COUNT` and `CPU-TEMP- $\{n\}$` MIB entries (Section 4.6.28). Hard drive temperatures are monitored via the `HDD-COUNT` and `HDD-TEMP- $\{n\}$` MIB entries (Section 4.6.30). MCS or an external monitoring system may poll these entries to detect overheating conditions.

Temperature monitoring depends on hardware support. Some hardware may not report HDD temperatures; in such cases, the `HDD-TEMP- $\{n\}$` response may be empty.

5.2 Storage Monitoring

Available storage space is tracked through the `TOTAL-STORAGE` and `REMAINING-STORAGE` MIB entries (Sections 4.6.21 and 4.6.22). The `REC` command checks for sufficient free space before scheduling a recording and rejects the request with an “Insufficient Drive Space” error if storage is insufficient.

Per-device storage is also available through the `DEVICE-COUNT` and `DEVICE-STORAGE- $\{n\}$` MIB entries for external devices.

5.3 Error Recovery

If the DR enters an error state, the `INI` command (Section 4.8.1) can be used to re-initialize the system without requiring a full reboot. The `--flush-log` and `--flush-data` flags provide additional control over what is reset during re-initialization.

When deployed as a `systemd` service, DROS2 is configured to restart automatically after an unexpected exit, with a 60-second delay between restart attempts.

5.4 Power

Standard computer power safety practices apply. No special power conditioning or UPS is required by the DR itself, although site-level power protection is recommended to avoid data loss from unexpected power failure during recording operations.

Section 6

References

6.1 LWA Documentation

1. MCS Common ICD (LWA Engineering Memo MCS0005)
2. MCS-NDP Interface Control Document
3. J. Craig, “Long Wavelength Array Station Architecture,” Ver. 2.0, LWA Memo 161, Feb. 26, 2009.
4. C. Wolfe, S. Ellingson, C. Patterson, “MCS Data Recorder Preliminary Design & Verification,” Ver. 1.0, LWA Memo 165, Aug. 26, 2009.
5. C. Wolfe, S. Ellingson, C. Patterson, “MCS-DR Storage Unit,” LWA Engineering Memo MCS0019, Sep. 23, 2009.

6.2 Standards

1. IEC 60027-2, “Letter symbols to be used in electrical technology — Part 2: Telecommunications and electronics,” Third Ed., 2005.

Software Components

Component	Description
DROS2	Main Data Recorder Operating Software. Provides all ICD-specified functionality.
Msender	Command-line utility to send MCS messages to and receive responses from the DR. Intended for operating the DR in the absence of MCS.
DataSource2	Simulated NDP data source for testing. Emulates NDP output in DRX, DRX8, TBT, TBS, and COR formats.
SpectrogramViewer	GUI utility for viewing spectrometer output files produced by DROS2.

<code>build_helper.py</code>	Build configuration utility for multi-DR deployments. Creates per-instance build directories with separate installation paths and service files.
------------------------------	--

Configuration Files

File	Description
<code>defaults_v2.cfg</code>	Main system configuration (Section 6.3).
<code>binary-mode</code>	Selects the active DROS2 operating mode (<code>Spectrometer</code> or <code>LiveBuffer</code>).
<code>dros.service</code>	<code>systemd</code> unit file for single-DR deployment.
<code>dros-drN.service</code>	<code>systemd</code> unit file for multi-DR deployment (one per DR instance).

6.3 Configuration Parameters

DROS2 configuration is stored in a key-value text file (`defaults_v2.cfg`). Table 6.3 lists the supported parameters in the order they appear in the configuration file.

Table 6.3: DR Configuration Parameters (`defaults_v2.cfg`)

Parameter	Type	Description
<code>MyReferenceDesignator</code>	string	Subsystem reference designator (e.g., <code>DR1</code>). Messages with non-matching designators (other than <code>ALL</code>) are ignored.
<code>SelfIp</code>	IP address	DR control interface IP address.
<code>TimeAuthority</code>	IP or hostname	NTP time server used for synchronization on boot and via the <code>SYN</code> command.
<code>SerialNumber</code>	string	Serial number identifying this DR instance.
<code>ArraySelect</code>	string	DRSU to select on startup. May be a 0-based index or a DRSU barcode.
<code>MessageInPort</code>	integer	UDP port for receiving MCS messages.
<code>MessageOutPort</code>	integer	UDP port for sending MCS responses.
<code>DataInIp</code>	IP address	IP address to bind for receiving NDP data frames. Defaults to the value of <code>SelfIp</code> .
<code>DataInPort</code>	integer	UDP port for receiving NDP data frames.

Appendix A

Spectrometer Data File Format

Spectrometer output files consist of a sequence of blocks. Each block contains a header followed by spectral data. For each “block” that produces spectra for all tuning/polarization combinations, one header structure and associated spectral data are written.

A.1 Header Structure

The header is defined by the following C structure (packed, no padding):

```
typedef struct __DrxSpectraHeader {
    uint32_t MAGIC1; // must always equal 0xCODECODE
    uint64_t timeTag0; // time tag of first frame in block
    uint16_t timeOffset; // time offset reported by NDP
    uint16_t decFactor; // decimation factor
    uint32_t freqCode[2]; // NDP frequency codes for each tuning
                        // index: 0..1 = Tuning 1..2
    uint32_t fills[4]; // fill counts per pol/tuning combination
                        // index: 0..3 = X0, Y0, X1, Y1
    uint8_t errors[4]; // error flag per pol/tuning combination
                        // index: 0..3 = X0, Y0, X1, Y1
    uint8_t beam; // beam number
    uint8_t stokes_format; // Stokes output product identifier
    uint8_t spec_version; // spectrometer version byte
    uint8_t reserved; // reserved flags (includes kurtosis)
    uint32_t nFreqs; // Transform Length (number of channels)
    uint32_t nInts; // Integration Count
    uint32_t satCount[4]; // saturation counts per pol/tuning
                        // index: 0..3 = X0, Y0, X1, Y1
    uint32_t MAGIC2; // must always equal 0xEDOCEDOC
} __attribute__((packed)) DrxSpectraHeader;
```

New in Version 2.4.1

The header structure has been extended from the version 1.4 definition:

- Added `stokes_format` field to identify the Stokes output product.
- Added `spec_version` byte.
- Added `reserved` flags byte (includes kurtosis flag).
- Added `satCount[4]` array for per-cell saturation counts.

A.2 Stokes Output Products

The `stokes_format` field in the header identifies the output product (Table A.1). The number of output products per block determines how many floating-point spectra follow the header.

Table A.1: Supported Stokes Products

Name	Hex Code	Products	Output Order
XXYY	0x09	2	$ X ^2, Y ^2$
CRCI	0x06	2	$\text{Re}(XY^*), \text{Im}(XY^*)$
IQUV	0xF0	4	I, Q, U, V
IV	0x90	2	I, V
XXCRCIYY	0x0F	4	$ X ^2, \text{Re}(XY^*), \text{Im}(XY^*), Y ^2$
I	0x10	1	I

A.3 Spectral Data Layout

Immediately following the header, the spectral data for each tuning/stream are written as 32-bit floating-point values. For a spectrometer with N_{streams} streams (typically 2 for DRX, corresponding to the two tunings), N_{products} output products, and N_{freqs} frequency channels, the data following each header consists of:

$$N_{\text{streams}} \times N_{\text{products}} \times N_{\text{freqs}} \text{ 32-bit floats}$$

The data are organized as: for each stream, for each product, N_{freqs} float values. The frequency channels are arranged with the DC component at the center (the output is “fftshifted”).

A.4 Resolution

The spectral and temporal resolution of the spectrometer output are determined by the DRX/DRX8 sample rate f_D , the Transform Length (number of channels) N_{chan} , and the Integration Count N_{int} .

The input sample rate is set by the DRX filter code in use:

$$f_D = \frac{f_s}{D}$$

where $f_s = 196$ MHz is the NDP sample clock and D is the decimation factor for the active filter code (see Appendix C, Tables C.2 and C.3).

The spectral resolution (channel width) is:

$$\Delta f = \frac{f_D}{N_{\text{chan}}}$$

The temporal resolution (integration time per output spectrum) is:

$$\Delta t = \frac{N_{\text{chan}} \times N_{\text{int}}}{f_D}$$

Note

Example: With $f_D = 19.6$ MHz (filter code 7, $D = 10$), $N_{\text{chan}} = 256$ channels, and $N_{\text{int}} = 4096$ integrations:

$\Delta f = 19.6 \text{ MHz} / 256 = 76.5625 \text{ kHz}$ per channel

$\Delta t = 256 \times 4096 / 19.6 \text{ MHz} \approx 53.5 \text{ ms}$ per spectrum

A.5 Error Handling

- If an error occurs in “cell” n : `fills[n]` is set to 0, all spectral data for that cell are set to 0.0, and `errors[n]` is set to 1.
- If cell n is not completely filled: `fills[n]` reflects the number of complete transforms performed. The spectral data are *not* normalized by `fills[n]`; the caller must perform this normalization.

A.6 Time Continuity

Generally, consecutive spectra blocks are continuous in time. NDP resets, time-tag rollover, or frame receive errors may cause non-integral time-tag steps between consecutive blocks. Only events giving rise to error flags are explicitly flagged.

Appendix B

Error Messages

When a command cannot be executed, the DR returns R in the R-RESPONSE field with an error message in the R-COMMENT field. Table B.1 lists the defined error messages.

Table B.1: DR Error Messages

Error Message	Description
Operation not permitted	Operations are scheduled or in progress that prevent execution of the requested command.
Invalid Name	The specified format name contains illegal characters.
Format Already Defined	The specified format name is already in use.
Invalid Size	The specified UDP packet payload size exceeds the maximum (8192 bytes accounting for headers).
Invalid Rate	The specified data rate exceeds system capabilities.
Already Up	Internal storage is already online.
Not Detected	Internal storage was not detected.
Cannot Start	File system information could not be determined on the specified device.
Already Down	Internal storage is already offline.
Invalid Storage ID	The specified storage ID does not exist.
File not found	The specified tag value does not refer to any file on internal storage.
Invalid Filename	The specified filename contains illegal characters.
Invalid Position	The requested start byte and length exceed the file size.
Invalid Range	The requested length exceeds response size limitations.
Not Scheduled	The specified tag does not refer to any scheduled or existing recording.

Error Message	Description
Already Stopped	The specified tag refers to an already-completed recording.
Invalid Time	The requested time frame is in the past or too far in the future.
Time Conflict: <Op>	The requested time frame overlaps with a scheduled operation. The first conflicting operation is listed in SCHEDULE-ENTRY-{n} format.
Unknown Format: <Fmt>	The specified data format is undefined or unusable.
Insufficient Drive Space	The storage required exceeds the remaining internal storage.
Component Not Available: <X>	Internal component X is unavailable, unusable, or malfunctioning.
Bad format: bad argument count	The command arguments could not be parsed; wrong number of fields supplied.
Unsupported channel count	The specified FFT length is outside the valid range (32–16384).
Unsupported integration count	The specified integration count is outside the valid range (384–24576).
Unsupported geometry	The product of Transform Length and Integration Count is not an integral multiple of the frame sample count.
Unknown Stokes output type	The specified Stokes product name in the metadata is not recognized.
Unsupported Stokes output type	The specified Stokes product is recognized but not supported.

Appendix C

Data Formats Reference

New in Version 2.4.1

This appendix is new in version 2.4.1. It provides a complete reference of all data formats supported by DROS2. The previously supported TBN and TBW formats have been removed; DRX8, TBT, TBS, and COR formats have been added.

C.1 Format Summary

All data formats consist of a frame header followed by sample data transmitted as UDP packets from NDP. The DR records complete frames (header + samples) to disk. Table C.1 summarizes the supported formats.

Table C.1: Data Format Summary

Format Type	Frame Size (bytes)	Samples/Frame	Sample Type
DRX	4128	4096	4-bit complex (PackedSample4)
DRX8	8224	4096	8-bit complex (PackedSample8)
TBT	$28 + 256 \times 32 = 8220$	256×32	4-bit complex
TBS (4 ch)	$28 + 256 \times 8 = 2076$	256×8	4-bit complex
TBS (8 ch)	$28 + 256 \times 16 = 4124$	256×16	4-bit complex
TBS (12 ch)	$28 + 256 \times 24 = 6172$	256×24	4-bit complex
COR	2336	288	64-bit complex (PackedSample64)

Note

TBT and TBS frame sizes depend on the number of stands (TBX_STAND_COUNT), which defaults to 256. The sizes shown above assume this default value. TBT uses $16 \times 2 = 32$ samples per stand (16 time samples, 2 polarizations). TBS uses $N_{\text{chan}} \times 2$ samples per stand, where N_{chan} is the number of frequency channels (4, 8, or 12).

C.2 DRX Formats

DRX carries 4-bit complex samples from the digital receiver. Four streams are produced per beam (2 tunings $\times$ 2 polarizations). The seven filter (decimation) settings listed in Table C.2 are available, corresponding to sample rates from 250 kHz to 19.6 MHz.

Table C.2: DRX Data Formats

Format Name	Dec. Factor	Rate (bytes/s)	Sample Rate
DRX_FILT_1	784	1,011,360	250 kHz
DRX_FILT_2	392	2,018,592	500 kHz
DRX_FILT_3	196	4,033,056	1 MHz
DRX_FILT_4	98	8,066,112	2 MHz
DRX_FILT_5	40	19,756,608	4.9 MHz
DRX_FILT_6	20	39,509,088	9.8 MHz
DRX_FILT_7	10	79,014,048	19.6 MHz
DEFAULT_DRX	–	79,014,048	19.6 MHz (default)

C.3 DRX8 Formats

DRX8 is identical to DRX except that samples are 8-bit complex instead of 4-bit complex, resulting in approximately double the data rate (Table C.3).

Table C.3: DRX8 Data Formats

Format Name	Dec. Factor	Rate (bytes/s)	Sample Rate
DRX8_FILT_1	784	2,006,656	250 kHz
DRX8_FILT_2	392	4,013,312	500 kHz
DRX8_FILT_3	196	8,026,624	1 MHz
DRX8_FILT_4	98	16,061,472	2 MHz
DRX8_FILT_5	40	39,351,840	4.9 MHz
DRX8_FILT_6	20	78,703,680	9.8 MHz
DRX8_FILT_7	10	157,415,584	19.6 MHz
DEFAULT_DRX8	–	157,415,584	19.6 MHz (default)

C.4 TBT Format

TBT (Transient Buffer — Triggered) carries time-domain voltage data for all stands at a single tuning (Table C.4). Each frame contains data for all stands with 16 time samples per channel (2 channels per stand, corresponding to 2 polarizations).

Table C.4: TBT Data Format

Format Name	Rate (bytes/s)
DEFAULT_TBT	117,440,512

C.5 TBS Format

TBS (Transient Buffer — Streaming) carries channelized voltage data for all stands (Table C.5). Three channel configurations are supported (4, 8, and 12 channels per polarization per stand); the default is 12 channels.

Table C.5: TBS Data Formats

Format Name	Channels	Rate (bytes/s)
TBS_FILT_7	4	49,670,376
TBS_FILT_8	8	98,670,824
TBS_FILT_9	12	147,671,272
DEFAULT_TBS	12	147,671,272 (default)

C.6 COR Format

COR (Correlator) carries correlation products (Table C.6). Each frame contains 288 64-bit complex samples.

Table C.6: COR Data Format

Format Name	Rate (bytes/s)
DEFAULT_COR	117,440,512